Compositional Neural Mechanics for Physical Prediction Beyond Visual Agreement

Anonymous CVPR submission

Abstract

A deformable digital twin should predict what an object will do under a new interaction, not only reproduce its observed images. These objectives differ: under displacement control, scaling material and attachment stiffness together can preserve every rendered view while changing reaction forces. We study a compositional representation that separates a shared neural bulk energy, compliant attachments, explicit contact and body loads, and material-attached Gaussian appearance. Compact energy decoders are pretrained on constitutive responses and frozen; scene adaptation estimates physical descriptors from observed positions. A conventional finite-element solver then predicts deformation and reactions under new commands or initial conditions. Controlled synthetic experiments test three distinct questions: transfer with known descriptors, refinement with unknown descriptors, and dynamic forecasting with shared initial appearance. On six withheld-composition objects, shared laws achieve 0.018 mm coordinate RMSE, versus 3.089 mm for a custom supervised direct predictor. An additional stretch observation reduces unknown-parameter prediction error from 0.339 to 0.080 mm, without establishing an adaptive-policy advantage. In a four-world drop study, the six forecasts available for both methods yield 1.146 versus 13.771 mm world-space vector RMS against native PhysTwin, and 18.73 versus 12.84 dB held-view foreground PSNR. We report missing forecasts, numerical-reference sensitivity and uncertainty miscalibration explicitly. The results support composition of learned local mechanics for controlled physical prediction; visual agreement alone remains insufficient evidence of physical accuracy.

1. Introduction

Images reveal where a surface is, but not necessarily why it is there. A soft object held between two grippers may deform because its bulk material is compliant, its attachments slip elastically, or both. A reconstruction can therefore render an observed motion convincingly while assigning the wrong response to the next manipulation. For action planning, a matched silhouette does not guarantee correct forces or rebound.

Recent Gaussian and continuum approaches have substantially improved the reconstruction and simulation of deformable scenes [4, 12, 14]. Learned constitutive representations also provide a route to sharing material knowledge across scenes [9, 10]. We build on these directions to ask a narrower question: can local learned material and interaction laws be composed into a reusable model whose physical predictions and rendered outputs are evaluated separately?

Our representation associates bulk properties with material coordinates, attachment laws with relations between material and prescribed tools, and collision geometry and gravity with the surrounding scene. Position commands specify how tools move; forces follow from the energy rather than being required as inputs. The physical state drives Gaussian centers and covariances, while appearance parameters remain independent of scene-specific mechanical fitting. This separation permits the same constitutive decoder to be reused across material combinations, geometries and interactions.

The distinction between appearance and mechanics is not merely a matter of metric choice. With no independent body load, multiplying every stiffness-bearing term in a position-controlled equilibrium problem by the same positive constant preserves its equilibrium configurations. Reactions change, but all cameras see the same geometry. Additional views cannot resolve this exact ambiguity. An informative action must change observability, for example through a known gravitational or inertial scale, or measured reactions.

We evaluate these issues through three controlled studies rather than pooling incompatible tasks. First, known-descriptor tests isolate forward transfer of learned local laws. Second, a small inverse problem estimates 15 continuous material and attachment descriptors and measures the benefit of one additional position observation. Third, timed drop experiments

compare frozen predictions against native PhysTwin using identical initial Gaussian appearance. Stronger compression, impact and bounce examples accompany the quantitative study, while numerical failures remain part of the record.

Our contributions are: (1) a concrete composition of compact conditional bulk and attachment energies with explicit loads, contact and Gaussian transport; (2) a scale-observability analysis connecting image ambiguity to physical response; and (3) a controlled evaluation separating transfer, adaptation, dynamics and appearance, including forecast coverage and reference sensitivity. The individual solver, invariant-energy and Gaussian-transport ingredients are established. Our evidence concerns their composition and the resulting prediction protocol, not a new general-purpose simulator or broad benchmark leadership.

2. Related Work

Physical neural scene representations. Gaussian splatting provides an explicit, efficiently rendered radiance-field representation [5]. Coupling visual representations to mechanics predates physical Gaussians: PAC-NeRF combines neural radiance fields with differentiable material point method (MPM) simulation to identify geometry and physical parameters from calibrated multi-view RGB video [7]. PhysGaussian reconstructs a static Gaussian scene and generates motion using supplied material and boundary settings; its continuum deformation map transports Gaussian covariances through the deformation gradient [12]. SpringGaus learns spring-mass dynamics from multi-view video through staged appearance, geometry and physical optimization [14]. PhysTwin uses sparse RGB-D interaction videos, shape priors and lifted image tracks to fit geometry and spring-mass physics, followed by Gaussian appearance optimization; it demonstrates future prediction and novel interactions [4]. These works establish both simulatable visual assets and separation of physical fitting from appearance.

Constitutive reuse and inverse mechanics. Learning material response within known governing equations is an established alternative to learning complete state transitions. NCLaw embeds structured constitutive networks in differentiable simulation and evaluates transfer across geometry, initial conditions and boundary conditions [9]. UniPhy learns shared latent-conditioned deformation-gradient projection and stress networks, pretrained using simulated constitutive supervision; at inference it freezes their weights and optimizes a material latent against observed particle trajectories [10]. Thus shared constitutive representations and scene-specific inverse adaptation already have direct precedent. Our formulation uses a scalar bulk-energy decoder, derives stress from that energy, and composes the bulk response with separately parameterized scene interfaces. The contribution concerns this composition and its evaluated predictive behavior, rather than priority for constitutive reuse. The elastic experiments also cover a narrower material class than UniPhy's plastic and fluid examples.

Generative priors and observed-motion reconstruction. PhysDreamer distills video-generation priors into material fields to synthesize responses to new interactions [13]. OmniPhysGS uses text-conditioned diffusion guidance to learn mixtures of constitutive experts for Gaussian particles [8]. Their generative supervision differs from fitting measured motion. PIDG instead uses time-dependent deformation, predicted velocity and stress, momentum residuals and optical-flow supervision for monocular dynamic reconstruction [3]. Its published formulation does not expose a query-force input. We distinguish reconstructing observed motion, generating plausible interactions and forecasting a reset system under changed loading; success on one task does not establish the others.

Numerics, systems and evaluation. DiffPD develops efficient differentiation through implicit projective dynamics, including contact models and inverse problems [2]. i-PhysGaussian couples Gaussian rendering to implicit Newton–GMRES MPM and evaluates time-step robustness [1]. Boba accelerates PhysTwin-style rollouts through compact spring-mass surrogates and shared-template batched execution [11]. These contributions motivate separating material representation from integration, inertia, contact and implementation effects. Deform360 provides real multi-view visuo-tactile data and evaluates per-episode, multi-episode and multi-object prediction [6]. Our controlled synthetic study uses known geometry and visible three-dimensional tracks, and compares future motion and Gaussian rendering under matched observations. It therefore evaluates a specific composed pipeline, with numerical-reference sensitivity and prediction coverage retained; it does not establish end-to-end video reconstruction, real-world material identification, calibrated posterior uncertainty or uncertainty-guided probing.

3. Compositional Neural Mechanics

3.1. State, inputs and outputs

Let $X \in \Omega_0 \subset \mathbb{R}^3$ denote a material coordinate and $x(X,t) = X + u(X,t)$ its current position. Its deforma-

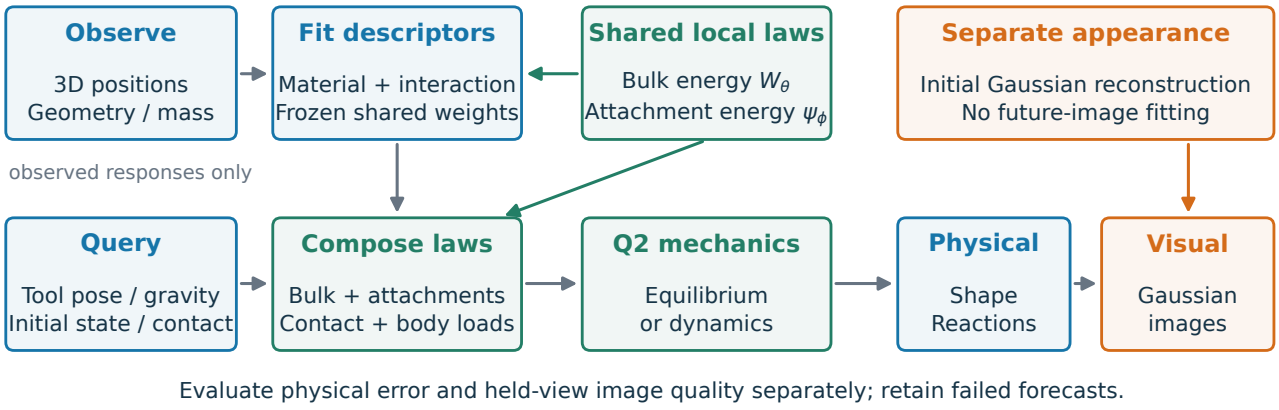


Figure 1. Representation and prediction pipeline. Local constitutive laws are trained once. Scene descriptors are fitted from observed positions with shared weights frozen. Prescribed tool poses, initial conditions, gravity and explicit contact geometry define a query; composed energies produce deformation and reactions through Q2 mechanics. The physical state transports a separately initialized Gaussian appearance model. Geometry and image metrics are evaluated independently on withheld responses. Geometry, mass and observation correspondences are supplied in the present experiments.

tion gradient is $F = \partial x/\partial X$. A scene has a material descriptor field $z_m(X)$, attachment descriptors z_b, known reference geometry, density, and explicit collision surfaces. A query specifies prescribed tool poses, gravitational acceleration and, for dynamics, initial positions and velocities. We predict x, deformation, reactions and images from requested cameras.

The input is therefore not an unknown force trajectory. A prescribed pose induces a displacement relative to the attached material; its reaction is an output. Gravity is an explicit body acceleration, and collision response depends on geometry and the predicted state. The present implementation supports isotropic hyperelasticity, bilateral compliant attachments and frictionless normal penalty contact. It does not discover arbitrary contact graphs from images. Rigid translation and rotation are contained in x; we do not train a separate rigid-motion carrier.

3.2. A compact shared bulk energy

Define $C = F^\top F$, $J = \det F > 0$, $I_1 = \operatorname{tr} C$, and $I_2 = ((\operatorname{tr} C)^2 - \operatorname{tr}(C^2))/2$. We use

$$q_1 = J^{-2/3} I_1 - 3, \quad q_2 = J^{-4/3} I_2 - 3, \quad v = J - \log J - 1. \tag{1}$$

For descriptor $z_m = (s, h, f)$, the energy density is

$$W_\theta(F; z_m) = e^s E_0 \sum_{j=1}^{7} a_j(h, f) b_j(F), \qquad E_0 = 12000\,\mathrm{Pa}, \tag{2}$$

where $a_j > 0$ are the outputs of a $2 \to 16 \to 7$ SiLU decoder followed by softplus. The basis is

$$b = [q_1/2, q_2/2, v, q_1^2, q_1^3, \phi_1(q_1), \phi_2(q_1)], \tag{3}$$

with $\phi_i(q) = \mathrm{softplus}(\alpha_i q) - \log 2 - \alpha_i q/2$ and learned positive α_i. The decoder and slopes together have 169 parameters. The external factor e^s preserves exact common stiffness scaling.

Stress $P = \partial W_\theta/\partial F$ and the tangent are obtained by automatic differentiation. Dependence on invariants gives objectivity and isotropy, and the basis has zero reference energy and stress. These properties do not establish general polyconvexity, unique equilibria or global numerical stability. The model represents a restricted family of elastic responses, without plasticity or material history.

Constitutive pretraining. We supervise energy and stress using 4,096 synthetic strain coupons from 18 descriptor combinations of compressible Mooney–Rivlin and Yeoh-type teachers. Training uses 6,000 Adam steps with batch size 256 and normalized stress MSE plus 0.1 times energy MSE. The two families share small-strain moduli but differ in nonlinear response. The weights are frozen before scene tests. This is constitutive supervision, not training from the benchmark's future images.

3.3. Attachments, contact and body loads

A prescribed tool pose maps a reference point to $T_b(X)$. Its attachment slip is $\delta_b = x(X) - T_b(X)$. Writing $r =$

$\|\delta_b\|^2/\ell^2$, with $\ell = 2\,\mathrm{mm}$, our conditional attachment energy is

$$\psi_\phi(\delta_b; K, \chi, f_b) = \tfrac{1}{2} K \ell^2 \left[r + 2\chi \sum_j a_j(\chi, f_b)\phi_j(r) \right]. \tag{4}$$

The positive-coefficient network has 179 parameters, including learned basis slopes and offsets. At $\chi = 0$ it is an exact quadratic spring. A Gaussian/taper footprint $w_b(X)$ is normalized over reference area so that K has units N/m. Footprint centers and widths can be inferred. Pretraining uses analytic polynomial and saturating attachment responses; details appear in the supplement.

Attachments are bilateral relations, not collision contact. Known plane or sphere geometry separately supplies a signed gap $g(x)$ and penalty energy

$$\Pi_c = \tfrac{1}{2} k_c \int_{\Gamma_0} [\min(g(x), 0)]^2 \, dA_0. \tag{5}$$

This permits finite penetration; there is no fitted restitution coefficient or friction law. The total potential is

$$\Pi(u; z, a) = \int_{\Omega_0} W_\theta(F, z_m)\, dV + \sum_b \int_{\Gamma_b} w_b \psi_\phi(\delta_b, z_b)\, dA + \Pi_c - \int_{\Omega_0} \rho g^\top x \, dV. \tag{6}$$

The scene thus composes distinct material, relational and world-dependent terms instead of treating every boundary effect as a single spatial stiffness field.

3.4. Solving and adapting a scene

We discretize u with tensor-product Q2 hexahedral elements. Static prediction solves $R = \nabla_u \Pi = 0$ with Newton iterations, sparse factorization and line search. Dynamic prediction uses consistent mass M and velocity Verlet for $M\ddot{u} = -\nabla_u \Pi$. Finite-mass spherical strikers receive equal-and-opposite normal contact forces. The main drop comparison is conservative; a separate viscous demonstration is not included in its aggregate metrics.

Scene adaptation minimizes observed position residuals while freezing both learned laws. The static inverse experiment fits 15 continuous descriptors, including material scale and shape, heterogeneity amplitude, two attachment stiffnesses and nonlinearities, and their footprint centers and widths. Categories and geometry remain known. Gauss–Newton sensitivities follow the equilibrium equation:

$$H \frac{\partial u}{\partial z} = -\frac{\partial R}{\partial z}, \qquad H = \frac{\partial R}{\partial u}. \tag{7}$$

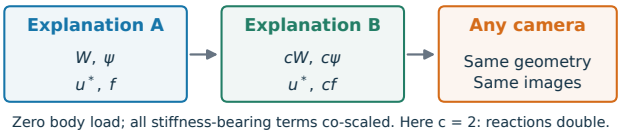


Figure 2. Visual agreement need not identify physical scale. Co-scaling bulk and attachment energies preserves position-controlled equilibria but changes reactions. An extra view cannot remove this particular ambiguity.

The dynamic experiment instead searches two material families and fits scale, shape and contact stiffness using a forward catalog followed by bounded-budget Powell optimization. It is a distinct derivative-free fitting procedure, not posterior sampling. Forecasts use the best saved finite candidates; most searches exhaust their budgets, so completed forecasts are not described as converged parameter estimates.

3.5. Material-attached Gaussian appearance

Following deformation-gradient transport [12], a canonical Gaussian bound at material point q, with rest offset $r = \mu_0 - X(q)$, becomes

$$\mu_t = x(q,t) + F(q,t)r, \qquad \Sigma_t = F(q,t)\Sigma_0 F(q,t)^\top. \tag{8}$$

Opacity and spherical-harmonic coefficients are frozen; the implementation does not rotate the latter. In the dynamic comparison, both methods start with the same learned Gaussian checkpoint, renderer and cameras. This controls appearance initialization while allowing the predicted mechanics and transport to differ. It does not compare independently reconstructed appearance quality.

4. What Can Images Identify?

Common-scale ambiguity. Consider prescribed-position equilibrium with no independent body load and scale all stiffness-bearing terms by $c > 0$. Then $\Pi_c(u, a) = c\Pi(u, a)$, so the minimizer and equilibrium sets are unchanged. At corresponding equilibria, reactions $f = \partial\Pi/\partial a$ and tangents scale by c. If geometry and appearance are identical, rendering any camera gives identical images. This statement does not require a unique equilibrium; it concerns corresponding solution branches or minimizers. If contact is present, its stiffness must also be scaled.

Our numerical counterexample doubles stiffness, obtains a scaled-equilibrium residual of 3.59×10^{-13} N and produces byte-identical images from all three tested views while doubling reactions. This is a controlled illustration of an exact symmetry, not evidence that all

visual inverse problems are unidentifiable. Known gravity or fixed-mass timed dynamics can break this scale symmetry; unknown density can introduce related ambiguities. Breaking one symmetry also does not guarantee globally identifiable parameters.

Actions and uncertainty. A new view can reveal previously occluded deformation but cannot reveal a mechanical quantity absent from all geometry. A new action can excite a different strain or attachment regime and reduce prediction error. We test the latter with an extra position observation. We distinguish its empirical benefit from the stronger claim that an uncertainty score selects better actions. In particular, local parameter uncertainty omits model discrepancy and can contract around an inaccurate explanation.

5. Experiments

5.1. Tasks and evaluation protocol

All experiments are synthetic, with known reference geometry and material correspondences. They isolate mechanics rather than the full image-to-geometry pipeline. We keep three protocols distinct: known-descriptor transfer, unknown-descriptor static refinement, and dynamic forecasting after calibration. No cross-task average is reported.

For static tests, coordinate RMSE averages squared errors over points and their three coordinates before taking the square root. Dynamic world error is point-vector RMS, using reference-area surface weights; shape error additionally removes one best-fitting rigid transform per frame. The latter distinguishes deformation from gross translation and rotation. Vector RMS equals $\sqrt{3}$ times coordinate RMSE only for identical samples, weights and aggregation without additional alignment.

We retain missing forecasts and distinguish complete-case paired summaries from whole-cohort coverage. Numerical-reference checks separately vary timestep and spatial resolution. An energy or penetration check alone is insufficient to establish spatial convergence. Supplementary material provides solver settings, per-case values, recovery history and videos.

5.2. Transfer with known descriptors

Training scenes contain 54 objects and six commands each. Tests comprise six withheld-composition objects and six geometry draws, with four new commands per object and 343 common material queries. The geometry draws use two unseen parameterizations of the same mapped-cuboid topology. All material and attachment descriptors are supplied, so this tests forward transfer rather than inverse identification. Predictions use resolution $n = 4$ and analytic references $n = 8$; the structured element layout is $(4n, n, n)$.

Table 1. Known-descriptor forward transfer. Coordinate RMSE in mm; six objects and four commands per object in each column. Direct is a custom 30k-step stable normalization control.

Method	Compositions	Geometries
Matched analytic	0.0164	0.0159
Shared laws	0.0181	0.0183
Custom direct, 30k	3.0888	3.2933

We compare shared laws with a matched analytic teacher, a Neo-Hookean alternative, and custom DeepSets direct predictors trained for 6,000 or 30,000 steps on the same 324 scene responses. These are supervised controls, not published graph-network baselines. The direct predictors receive matched descriptors and frozen-law fingerprints. Their reported stable controls repair zero-variance input normalization with physical units while keeping weights fixed; the supplement records the original failures and recovery protocol.

Table 1 shows that shared local laws remain close to the matched analytic model under these changes, while the direct predictor has larger errors. Figure 3 includes both training budgets and the simpler constitutive control. This supports reuse within the represented law family; it does not establish transfer to arbitrary unseen materials or interaction types. The small analytic-model error also includes discretization differences.

5.3. Refining unknown descriptors with an action

A separate experiment uses six fresh objects, 26 observed/scored points, 50 μm position noise and two withheld commands per object. Geometry, categorical material/interface families, heterogeneity pattern, gravity and density are known; 15 continuous descriptors are fitted. Inference uses $n = 2$ and the analytic reference $n = 4$. No reactions are observed.

An extra stretch observation lowers mean held-command position coordinate RMSE from 0.3395 to 0.0802 mm and reaction coordinate RMSE from 0.0961 to 0.0292 N. Thus position observations can improve predictions of an unobserved physical output when the setup provides an independent scale. The variance-based policy selects stretch for all six objects, however, and exactly ties the fixed stretch policy. It has no demonstrated advantage over that strong fixed choice. Of 48 fits across the declared variants, 41 converge and

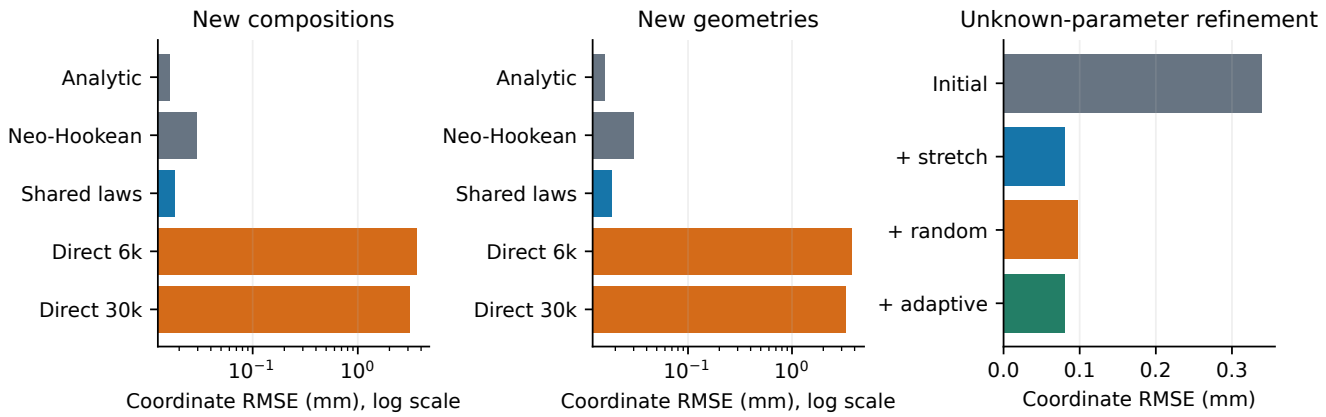


Figure 3. Forward transfer and additional observations. Left and center: known-descriptor coordinate RMSE for new compositions and geometry draws, on logarithmic axes. Right: unknown-descriptor prediction error after one additional observation. Adaptive selection chooses the same stretch on every object and ties the fixed stretch policy. Direct controls use frozen weights with the documented stable normalization.

seven incomplete fits are retained; all stretch fits converge.

Moreover, nominal 90% position-interval coverage falls from 97.0% initially to 33.0% after stretch. Better point predictions coexist with worse uncertainty calibration. We therefore report the additional-observation result as prediction refinement, not as calibrated active system identification.

5.4. Dynamic prediction against native PhysTwin

The fresh cohort has four worlds: two Mooney–Rivlin and two Yeoh-type materials with randomized scale, shape and contact stiffness. Each uses a $12 \times 6 \times 6$ cm block of mass 0.27 kg under gravity. Calibration observes a 10 mm drop for 0.20 s; forecasts predict 40 and 80 mm drops for 0.30 s at 100 Hz. This is an out-of-calibration load-amplitude test within the same contact class, not unknown-interaction-type generalization. Calibration uses noiseless visible 3D tracks. Ninety-eight unique surface points and reference-area weights are shared for scoring.

We run the authors' native PhysTwin implementation [4], retaining its 125-node spring graph and native fitting pipeline. A global recipe of at most 100 CMA generations, retaining native early stopping, and 1,000 optimization epochs is chosen on development calibration only and frozen before fresh world draws. Our continuum model uses $n = 3$ and 50 μs timesteps; the primary reference uses $n = 4$ and 25 μs. Native and continuum parameterizations, mass distributions, damping and contact models differ, so this is a configuration-level comparison rather than an isolated architecture ablation.

Our method completes all eight forecasts. Native PhysTwin completes six; one world's graph initialization fails with an empty graph before its two forecasts. These are recorded as unavailable, not zero error or omitted successes. On the paired six forecasts from three worlds, world-space vector RMS is 1.146 versus 13.771 mm and rigid-aligned shape RMS is 1.085 versus 2.936 mm (Table 2). The world-space gap is substantially larger than the shape gap, demonstrating why rigid-aligned scoring matters.

Table 2. Fresh dynamic cohort. Error means condition on the six forecasts available to both methods. No complete eight-forecast paired mean exists. Held-view means use the same three worlds.

Measure	Ours	Native PhysTwin
Forecasts completed	8/8	6/8
World vector RMS (mm)↓	1.146	13.771
Shape vector RMS (mm)↓	1.085	2.936
Centroid error (mm)↓	0.346	13.412
Held FG PSNR (dB)↑	18.73	12.84
Held FG SSIM↑	0.747	0.577
Held silhouette IoU↑	0.956	0.795
Visible depth error (mm)↓	5.478	14.557

Appearance under predicted dynamics. Both models use the same initial appearance checkpoint, with 10,059 surviving Gaussians after training on three initial views. We freeze appearance and evaluate three observed and two withheld cameras over all 31 forecast frames. There are 2,480 method/image slots: 2,170 scored and 310 unavailable native slots. The paired held-view comparison uses 372 of 496 possible

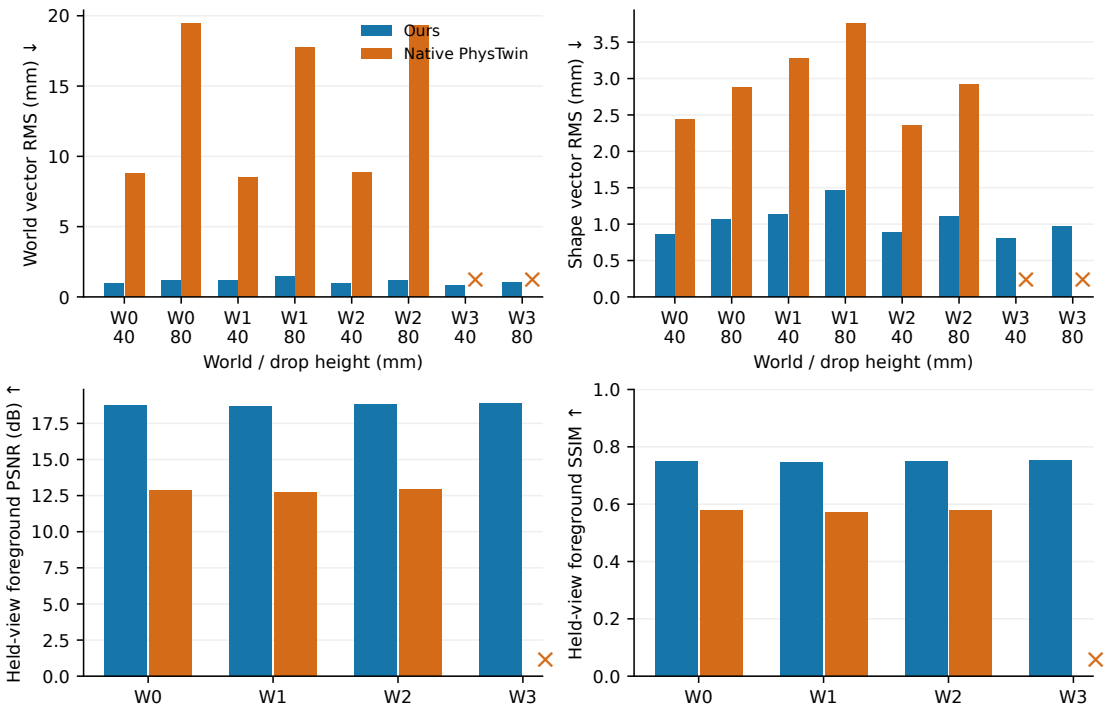


Figure 4. All declared dynamic cases. Top: world and rigid-aligned shape errors for both forecast heights in each world. Bottom: held-view image metrics by world. Crosses mark unavailable native predictions in world 3; they are not zero-valued measurements. Shared appearance initialization makes the image comparison a test of predicted motion plus Gaussian transport.

images per method, averaged equally over the same three worlds. Initial rendered RGB is byte-identical for paired queries. Held-view foreground PSNR and SSIM favor our model (Table 2), while Fig. 5 shows the first declared world's 80 mm forecast. These results concern physical trajectories and different transport rules with shared appearance, not superior radiance reconstruction.

Reference and optimizer sensitivity. All 20 reference-check simulations complete and pass individual energy, Jacobian and penetration checks. All four timestep comparisons pass, but all four declared spatial comparisons fail. In a post-hoc subset of two 80 mm forecasts, changing the reference from $n = 4$ to $n = 6$ changes our world errors from 1.180 to 1.690 mm and from 1.485 to 1.660 mm; the native errors change from 19.424 to 18.779 mm and from 17.720 to 17.005 mm. A further half timestep has little effect, and the ordering remains unchanged. This supports the ordering on this subset but does not qualify the primary reference as spatially converged. Predictions are not refitted to the refined targets.

Only one of 16 final dynamic optimizer searches meets its convergence criterion; 15 exhaust their declared budgets. Forecast coverage therefore measures saved finite trajectories, not certainty that the best physical parameters were recovered. A development-only removal of fitted native damping reduces world error from 9.518 to 6.147 mm while worsening shape error from 2.304 to 2.761 mm. Damping explains part of the gap, and our conservative prior matches the conservative teacher. Native graph mass discretization also changes rest inertia. These controls limit a claim of architecture-only superiority.

5.5. Stronger deformation and contact examples

The accompanying examples extend beyond barely visible motion. Prescribed 30% tool closure produces 22.7% material shortening; six mesh, timestep and quadrature configurations qualify that compression cycle. A finite 80 g sphere of radius 25 mm striking at 1 m/s produces 13.6% central compression and rebound. A separate viscous 40 mm drop demonstrates bounce and settling over 0.75 s. The impact and viscous examples are demonstrations, not spatially qualified comparative inference results. Abrupt-release failures at stronger deformation are retained. These examples demonstrate supported interactions beyond the quantitative forecast cohort.

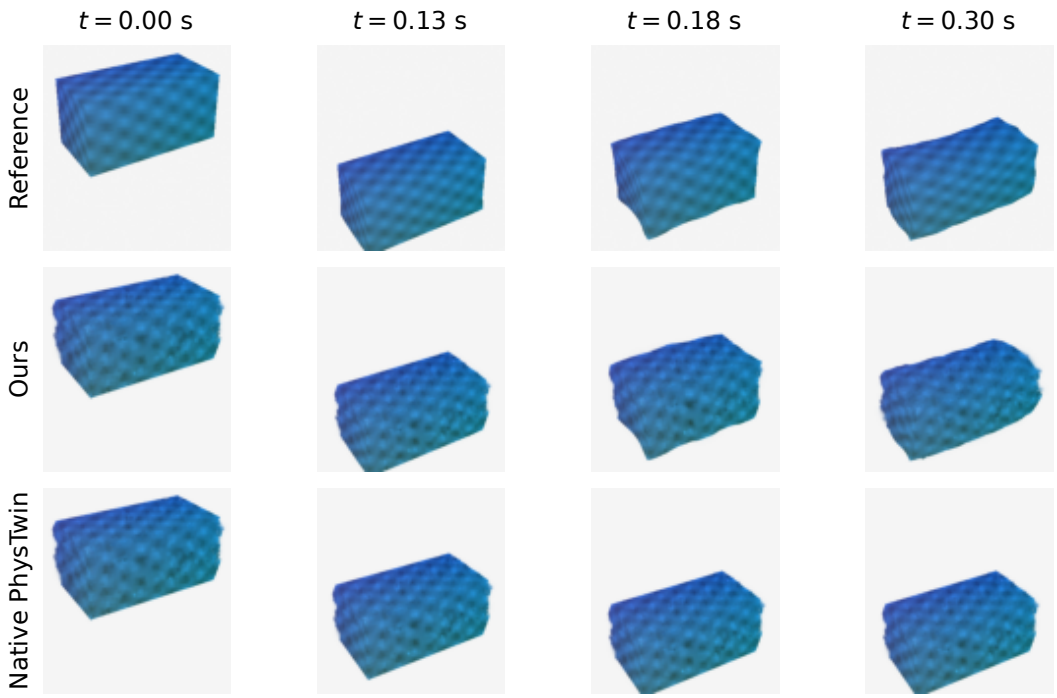


Figure 5. Actual reference versus predicted appearance. First declared world, 80 mm forecast, first held camera. The same crop is computed across all methods and all 31 frames; RGB values are unchanged. Columns show 0, 0.13, 0.18 and 0.30 s. This selection is a fixed case illustration, not an aggregate result. Supplementary videos include all eight forecasts, including the unavailable native case.

6. Discussion and Limitations

The strongest supported result is that reusable local laws can predict held responses accurately within a controlled family, and can improve physical trajectories and rendered forecasts relative to a configured native baseline in a small synthetic cohort. The scale example explains why images and physics must be evaluated separately. Additional observations help point prediction, but our uncertainty model and action selection do not yet justify reliable autonomous probing.

The experiments supply geometry, correspondences, density and substantial family knowledge. They do not establish image-only identification, performance on real captures, arbitrary unknown boundary conditions, or frictional multi-object contact. Constitutive teachers overlap the modeled family, and the dynamic study has only four worlds. Its forecasts are correlated within worlds; we do not treat hundreds of rendered frames as independent physical trials. The native baseline has two missing forecasts and differs in important physical priors. Reference spatial convergence remains unresolved for the main drop cohort.

A broader evaluation should add independent material families, real observations, stronger native baselines under matched dissipative assumptions, and spatially qualified references. For active identification, views should reveal uncertain geometry and interactions should distinguish physical responses. Calibrated action selection remains open.

7. Conclusion

We presented a compositional neural mechanics representation that separates shared local bulk and attachment laws from scene loads, contact and appearance. Frozen constitutive decoders support position-driven forward prediction and scene-specific descriptor fitting. Controlled transfer, refinement and dynamic experiments show useful prediction gains while exposing the limits of visual agreement, local uncertainty and finite-resolution references. A physically useful digital twin must be judged by its response to withheld interactions as well as by the images it renders.

LLM Usage Statement

OpenAI Codex was used extensively to assist with implementation, experiment orchestration, literature discovery, analysis checks, manuscript drafting and programmatic figure generation. Numerical results are drawn from saved experiment artifacts; scientific figures and qualitative panels are generated from those artifacts rather than synthesized as experimental evidence. This assistance does not replace author responsibility for validating the methods, references, results and final submission.

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